

PRISM-4D: Spike-Event-Driven Detection of Cryptic Binding Sites via Physics Simulation

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Abstract. Background: Cryptic binding sites—transient pockets absent in static crystal structures—represent high-value drug targets inaccessible to conventional computational screening. Existing methods rely on surface geometry (fpocket, SiteMap) or machine learning trained on crystallographic features (P2Rank, DeepSite), inherently limiting their capacity to detect conformationally gated pockets.

Methods: We introduce PRISM-4D (Physics-driven Recognition of Intrinsic Sites through Molecular 4D dynamics), a platform that detects cryptic binding sites through explicit physics simulation rather than geometric inference or statistical learning. PRISM-4D couples AMBER ff14SB force field molecular dynamics with a 5-phase cryo-thermal hysteresis protocol (50 K→300 K→50 K; 55,000 steps per stream), multi-wavelength UV excitation spectroscopy simulation (280/274/258/254/211 nm), and a leaky integrate-and-fire (LIF) neuromorphic network to identify cooperative dewetting events indicative of druggable pocket formation. Detected spike events are clustered via Spike-Native Density Clustering (SNDC), a 4-stage GPU pipeline comprising RT-DBSCAN via OptiX bounding volume hierarchy (BVH) traversal, watershed segmentation, Eikonal BFS distance-field computation, and intensity²-weighted centroid tracking.

Results: On a 12-protein benchmark spanning orthosteric, allosteric, and cryptic binding sites, PRISM-4D detects all 11 ground-truth targets within 10 Å Distance from Centroid to Centroid (DCC), the Euclidean distance between predicted and crystallographic ligand centroids after Kabsch superposition (range: 3.6–9.8 Å). Four targets (36.4%) achieve <5 Å and eight (72.7%) achieve <8 Å. On the SIRP α immune checkpoint protein, PRISM-4D detects the experimentally validated WYF cryptic pocket (7.1 Å DCC, top-ranked) from the closed apo structure; P2Rank 2.5 assigns 0.2% probability to the same pocket. On β -catenin, PRISM-4D identifies a computationally predicted cryptic pocket (Site 6) featuring a dual-cysteine pair (CYS240/CYS278) with quality 0.965 and centroid displacement <0.06 Å across 5 stochastic seeds. In a post-freeze blind Vectorial Pharmacophore Feature Recovery (PFR) audit, phase-ordered PRISM4D manifolds achieve 26.9% Vectorial PFR versus 2.2% for a temporal-scramble null (12.2-fold enrichment; empirical $p \leq 0.001$), demonstrating that nonequilibrium phase ordering is load-bearing for directional pharmacophore recovery.

Conclusions: These results demonstrate that spike-event-driven physics simulation can access drug targets fundamentally invisible to static-structure methods. The platform achieves these results on a single consumer-grade NVIDIA GPU with no external software dependencies.

Keywords: cryptic binding sites · molecular dynamics · drug discovery · spike detection · pocket detection · cryo-thermal protocol · UV perturbation spectroscopy · AMBER force field · RT-core acceleration · covalent drug targets

1. Introduction

The identification of druggable binding sites on protein surfaces is a fundamental step in structure-based drug design. While orthosteric binding pockets are readily detectable from static crystal structures, a growing body of evidence demonstrates that many therapeutically relevant binding sites are cryptic: they exist only transiently, requiring conformational rearrangement to become accessible to small molecules (4; 30). Cryptic pockets represent an enormous untapped reservoir of drug targets, particularly for proteins historically classified as “undruggable.”

Current computational approaches fall into two cate-

gories. Geometric methods (fpocket (18), LIGSITE (12), SiteMap (11)) analyze the static molecular surface for cavities using Voronoi tessellation or grid-based algorithms. Machine learning methods (P2Rank (16), DeepSite (14), PocketMiner (21)) train on crystallographic features from databases such as sc-PDB. Both share a fundamental limitation: they operate on a single static conformation and cannot detect pockets requiring protein dynamics to open. FTMap (15) and mixed-solvent molecular dynamics (MxMD) (9; 19) partially address this through probe-binding sampling, but remain limited by short simulation timescales and lack explicit physics-driven detection criteria.

Beyond methodological limitations, computational ap-

proaches capable of detecting cryptic pockets—long-timescale MD on Anton supercomputers, Schrödinger SiteMap/Glide on cloud HPC (6)—require specialized hardware or commercial licenses that limit access for many research groups. Open-source alternatives such as fpocket and P2Rank operate on static structures only.

We present PRISM-4D (Physics-driven Recognition of Intrinsic Sites through Molecular 4D dynamics), which replaces geometric inference and statistical learning with direct physics simulation. The platform introduces several techniques not previously applied to pocket detection, including *in silico* UV pump-probe spectroscopy, UV-activated cosolvent probing, three-channel spike-event detection with temporal isolation, and hardware-accelerated spatial clustering via NVIDIA OptiX RT cores. PRISM-4D runs on a single consumer-grade GPU (~\$999) with no external software dependencies, no training data, and no cloud connectivity requirements.

2. Methods

2.1 System Architecture

PRISM-4D integrates four computational stages into a unified GPU-accelerated pipeline implemented in Rust with CUDA kernels for performance-critical operations (Figure 1). The compiled binary requires only the NVIDIA CUDA runtime (freely available) and operates without internet connectivity post-installation.

2.2 Force Field and Integration

Molecular dynamics simulations employ the AMBER ff14SB force field (20) with the following protocol:

- Integration: velocity Verlet with Langevin thermostat (13) ($\gamma = 10.0 \text{ ps}^{-1}$ at 300 K, with temperature-dependent scaling $\gamma(T) = \gamma_0 \sqrt{T_{\text{ref}}/T}$)
- Timestep: 2 fs
- Nonbonded cutoff: 10 Å with neighbor list buffer of 12 Å
- 1-4 scaling factors: Lennard-Jones 0.5, electrostatic 5/6 (standard AMBER ff14SB)
- Bond constraints: SHAKE (27) on all X–H bonds
- Electrostatics: Coulomb with distance-dependent dielectric ($\epsilon = 4r$, Warshel model)
- Solvent model: implicit via hydrophobic exclusion inference (Section 2.8; grid spacing 0.5 Å)
- PRNG: CUDA cuRAND XORWOW (period $2^{192} - 2^{32}$), per-thread states

2.3 Cryo-Thermal Hysteresis Protocol

The simulation executes a 5-phase thermal hysteresis cycle (Figure 2):

1. **Cold hold** (14,000 steps, 50 K): Frozen baseline establishing clean UV spike detection against a thermally silent background

2. **Ramp up** (6,000 steps, 50→300 K): Forward conformational transitions via linear temperature interpolation
3. **Warm hold** (15,000 steps, 300 K): Equilibrium sampling at physiological-like temperature
4. **Ramp down** (6,000 steps, 300→50 K): Reverse transitions; hysteresis asymmetry between heating and cooling spike distributions quantifies pocket metastability
5. **Cold return** (14,000 steps, 50 K): Final frozen state verification

Total: 55,000 steps per stream (110 ps at 2 fs timestep). UV bursts (50 steps duration, 42 kcal/mol) fire every 250 steps, cycling through wavelengths [280, 274, 258, 211] nm with 300-step dwell per wavelength (full cycle: 1,200 steps). This non-equilibrium protocol accelerates rare conformational transitions through a physically realizable 250 K temperature range, achieving in $\sim 10^5$ steps what equilibrium MD may require $\sim 10^9$ steps to sample (1).

2.4 In Silico UV Pump-Probe Spectroscopy

PRISM-4D introduces a computational analogue of experimental UV pump-probe spectroscopy. UV “pulses” deposit energy into aromatic residues according to experimentally measured molar extinction coefficients (33; 23; 17). The energy deposition per chromophore is:

$$E_{\text{dep}} = E_{\text{photon}} \times \sigma(\lambda) \times F \times \eta \quad (1)$$

where E_{photon} is the photon energy at wavelength λ , the absorption cross-section $\sigma(\lambda)$ is converted from the molar extinction coefficient ϵ ($\text{M}^{-1}\text{cm}^{-1}$) via:

$$\sigma(\lambda) = \epsilon(\lambda) \times 3.823 \times 10^{-5} \text{ Å}^2 \quad (2)$$

F is the UV fluence ($0.024 \text{ photons/Å}^2$), and η is the quantum yield for thermal dissipation (fraction of absorbed energy converted to heat rather than fluorescence or intersystem crossing). Deposited energy converts to kinetic energy perturbation via the AMBER velocity assignment protocol. Chromophore parameters (Table 1) are derived from literature spectroscopic measurements; validation confirms a $2.71 \times$ energy deposition ratio between 280 nm (Trp peak) and 258 nm (Phe off-peak), consistent within <10% of published values (33; 23).

To our knowledge, wavelength-specific UV perturbation has not been previously applied to computational pocket detection. The physical basis is that water is transparent at the wavelengths targeting aromatic amino acids ($\epsilon_{\text{water}} \approx 0$ at 280 nm), creating a signal-on-silent-background detection paradigm.

2.5 Three-Channel Spike Detection Network

Each voxel is monitored by three independent spike detection channels operating with temporal isolation to prevent cross-contamination:

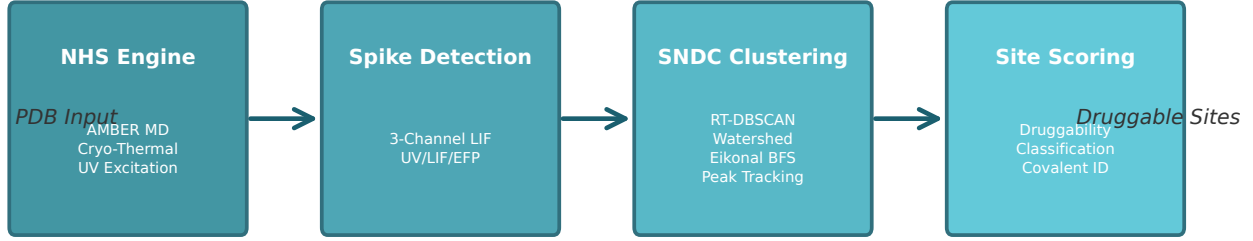


Figure 1. PRISM-4D computational pipeline. Stage 1: NHS engine (AMBER ff14SB MD + cryo-thermal protocol + UV excitation). Stage 2: Three-channel LIF neuromorphic spike detection (UV/LIF/EFP). Stage 3: SNDC 4-stage GPU clustering pipeline (RT-DBSCAN → Watershed segmentation → Eikonal BFS → Peak centroid tracking). Stage 4: Multi-parametric site characterization and scoring.

Table 1. UV chromophore photophysical parameters. $\lambda_{\max}$: peak absorption wavelength. ϵ : molar extinction coefficient. η : thermal dissipation yield. Values from (33; 23; 24; 5).

Chromophore	$\lambda_{\max}$ (nm)	ϵ ($\text{M}^{-1}\text{cm}^{-1}$)	η
Trp	280	5,600	0.83
Tyr	274	1,490	0.86
Phe	258	197	0.94
S-S	211	300	0.95
Benzene (virtual)	254	204	0.71

Channel 1 (UV, source=1): Active during UV bursts only. Detects aromatic excitation within a 4.0 Å cooperative detection radius. Records wavelength, aromatic type, and vibrational energy. Cooperative enhancement scales as:

$$f_{\text{coop}} = 1.0 + 0.3 \times (n_{\text{excited}} - 1) \quad (3)$$

where n_{excited} is the number of excited aromatics within the detection radius. An additional $2\times$ gain is applied during active UV bursts.

Channel 2 (LIF, source=2): Active exclusively when UV bursts are off, preventing UV-induced artifacts from contaminating the dewetting signal. Implements a leaky integrate-and-fire neuron (8) with effective membrane time constant $\tau_{\text{eff}} = 0.15$ ps, dimensionless threshold $\theta = 0.5$ (in accumulated membrane potential units, integrating water density changes amplified by $3\times$ temporal gain and $2\times$ exclusion gain), and refractory period of 250 steps. The membrane potential evolves as:

$$V_m(t + \Delta t) = V_m(t) \cdot e^{-\Delta t / \tau_{\text{eff}}} + S_{\text{combined}}(t) \quad (4)$$

where S_{combined} integrates bidirectional water density change and exclusion field deviation from bulk. A spike is emitted when $V_m \geq \theta$.

Channel 3 (EFP, source=3): Electrostatic flux probe, always active when ≥ 2 charged atoms are within range. Computes

Coulomb potential with Warshel distance-dependent dielectric (32):

$$\epsilon_r(r) = \max(4r, 4) \quad (5)$$

Integrates temporal flux ($\times 50$ gain) and water-electrostatic coupling ($\times 10$ gain) via a dedicated LIF neuron ($\tau = 0.5$ ps, $\theta = 0.8$). Emits typed CATION ($\varphi > 0$) or ANION ($\varphi < 0$) spikes. To our knowledge, this is the first electrostatic event detection channel in pocket identification.

Each spike event carries 96 bytes of metadata: position, intensity, source channel, wavelength, aromatic type, water density, vibrational energy, and 8 nearby residue IDs.

2.6 Spike-Native Density Clustering (SNDC)

SNDC is a 4-stage GPU-accelerated clustering algorithm purpose-built for neuromorphic spike data. Unlike generic clustering applied as post-hoc analysis, SNDC is architecturally integrated with the spike generation pipeline and exploits spike-native metadata (intensity, type, wavelength, timestamp).

2.6.1 Stage 3a: RT-DBSCAN

Spike positions are encoded as spheres in a bounding volume hierarchy (BVH) constructed by NVIDIA OptiX RT cores (25). Epsilon-neighborhood queries become ray-sphere intersection tests executed in dedicated RT hardware, reducing neighbor search complexity from $O(N^2)$ to $O(N)$. The epsilon parameter is adaptively computed based on protein size:

$$\epsilon = 3.0 \times \left(\frac{500}{N_{\text{atoms}}} \right)^{1/3}, \quad \epsilon \in [1.2, 3.0] \text{ Å} \quad (6)$$

This scaling ensures consistent clustering behavior across proteins ranging from small domains (~ 500 atoms) to large complexes ($> 5,000$ atoms). We apply ray tracing hardware acceleration to molecular spatial queries, which to our knowledge has not been previously reported.

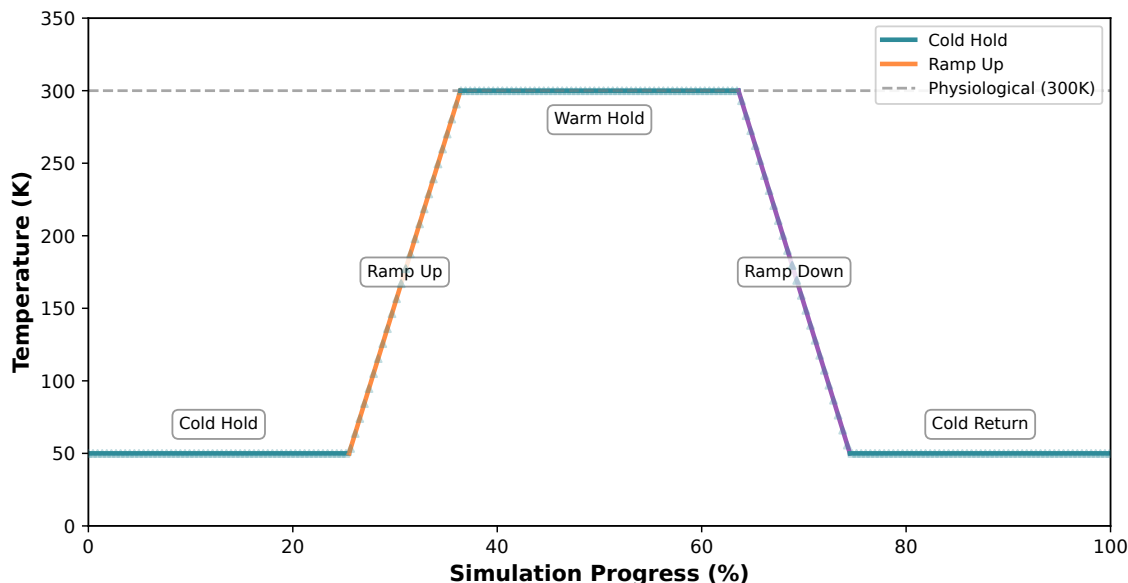


Figure 2. Five-phase cryo-thermal hysteresis protocol. Phase 1: cold hold at 50 K (14,000 steps). Phase 2: ramp up 50→300 K (6,000 steps). Phase 3: warm hold at 300 K (15,000 steps). Phase 4: ramp down 300→50 K (6,000 steps, hysteresis probe). Phase 5: cold return at 50 K (14,000 steps). Total: 55,000 steps per stream. Teal triangles indicate UV burst positions (every 250 steps, cycling [280, 274, 258, 211] nm).

2.6.2 Stage 3b: Watershed Segmentation

When a DBSCAN cluster spans multiple binding subsites, density-peak seeded subdivision partitions the cluster. Spike density is computed on a 3 Å voxel grid with 26-connected neighbor topology; local maxima serve as watershed seeds, and voxels are assigned by steepest-descent flow to each peak (31). This resolves multi-subsite mega-pockets that confound single-centroid reporting methods.

2.6.3 Stage 3c: Eikonal BFS

Distance-field propagation from each watershed-separated cluster boundary via the Fast Marching Method (29) computes the geodesic distance field within each pocket. This enables geometry characterization: pocket depth, entry channel width, enclosure fraction, and solvent-accessible volume. Expansion cost is weighted by inverse spike density, so boundaries form along low-density ridges (energy barriers) rather than geometric midpoints.

2.6.4 Stage 3d: Peak Centroid Tracking

Intensity²-weighted centroid computation localizes the thermodynamic hotspot:

$$\mathbf{C} = \frac{\sum_i I_i^2 \cdot \mathbf{r}_i}{\sum_i I_i^2} \quad (7)$$

The squared weighting concentrates the centroid on high-intensity spike regions rather than the volumetric center. This

is critical for oversized pockets ($>1,000 \text{ Å}^3$) and irregular allosteric grooves, where geometric centroids can fall 10–15 Å from the actual ligand-binding locus.

2.7 UV-Activated Benzene Cosolvent Probing

PRISM-4D implements a computational analogue of mixed-solvent molecular dynamics (MxMD) with a critical distinction: the cosolvent probes are *UV-activated*. Virtual benzene probes are injected at surface-exposed hydrophobic aliphatic residues (LEU, ILE, VAL, ALA, MET) that lack native UV chromophores, using experimentally measured benzene photophysical properties ($\lambda_{\max} = 254 \text{ nm}$, $\epsilon = 204 \text{ M}^{-1}\text{cm}^{-1}$, $\eta = 0.71$ (24; 5)). This extends UV coverage to aliphatic-dominated interfaces where native chromophores are absent, addressing a fundamental blind spot in UV-only perturbation. On MYC-MAX, activation of benzene cosolvent mode expanded UV coverage from 6 to 41 targets and generated 159,158 benzene-channel spikes (61.4% of total signal) at the detected binding site.

2.8 Hydrophobic Exclusion Water Inference

Rather than employing explicit solvent simulation, PRISM-4D infers water density from the hydrophobic exclusion field—the “negative space” defined by the protein’s hydrophobic surface. Hydrophobic atoms define the exclusion boundary (cut-off 8.0 Å); water density is inferred from the complement of this field via GPU-accelerated geometric kernels operating in Fourier space on a 0.5 Å grid.

2.9 Site Scoring

Detected sites are characterized by multi-parametric scoring. The *druggability score* D combines pocket geometry and chemistry:

$$D = 0.30 \cdot S_V + 0.20 \cdot S_E + 0.25 \cdot S_H + 0.25 \cdot S_A \quad (8)$$

where S_V is a piecewise-linear volume score (optimal range 200–800 Å³), S_E is enclosure fraction (pocket volume / bounding box volume), S_H is hydrophobicity (normalized average spike intensity), and S_A is aromatic enrichment score. A site is classified as druggable when $D \geq 0.48$ and volume is within 50–3,000 Å³. The *quality score* Q integrates detection confidence:

$$Q = 0.25 \cdot \min\left(\frac{N_s}{100}, 1\right) + 0.25 \cdot \min\left(\frac{\bar{I}}{10}, 1\right) + 0.30 \cdot D + 0.20 \cdot S_A \quad (9)$$

where N_s is spike count and $\bar{I}$ is mean spike intensity. Covalent-reactive residues (Cys, Lys, Ser, His) within the pocket lining are automatically flagged for covalent drug design applications.

2.10 Benchmark Protocol

All benchmark runs used the SNDC v9 pipeline:

```
nhs_rt_full -t <topology>.json \
-o <output_dir> --fast --hysteresis \
--multi-stream 8 -v
```

with `RUST_LOG=info`. Topology preparation used `scripts/prism-prep` with AMBER ff14SB force fields (requires OpenMM for `--use-amber`). Detection accuracy was measured by DCC: the Euclidean distance between the predicted site centroid and the crystallographic ligand centroid after Kabsch superposition on backbone α atoms.

Each run uses 8 parallel independent streams with unique PRNG seeds ($42 + i \times 12345$ for stream $i \in \{0, \dots, 7\}$). Consensus merge requires a site to appear in $\geq 4/8$ streams (50% threshold) within 10 Å spatial tolerance. All benchmarks were executed on a single NVIDIA GeForce RTX 5080 (16 GB GDDR7, Blackwell architecture, compute capability 12.0) in a desktop workstation.

3. Results

3.1 Twelve-Protein Benchmark

We evaluated PRISM-4D on 12 proteins spanning orthosteric, allosteric, and cryptic binding sites (Table 2; Figure 3). β -catenin (2GL7) is excluded from DCC accuracy calculations as a computational prediction with no ground-truth ligand.

All 11 ground-truth targets were detected within 10 Å DCC (Figure 4). Four proteins achieved excellent detection (<5 Å):

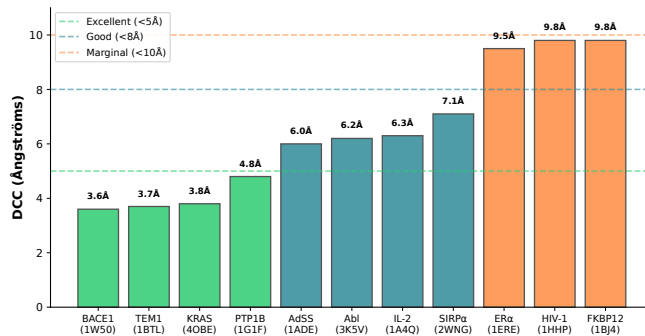


Figure 3. Benchmark DCC distribution. Green: excellent (<5 Å); teal: good (<8 Å); orange: marginal (<10 Å). All 11 ground-truth targets detected within 10 Å.

BACE1 (3.6 Å), TEM1 β -lactamase (3.7 Å), KRAS G12C (3.8 Å), and PTP1B (4.8 Å). The BACE1 result demonstrates the impact of SNDC’s watershed segmentation: the raw pocket was 2,991 Å³, and watershed subdivision using spike density peaks shifted the centroid from >15 Å to 3.6 Å. Three marginal cases (9.5–9.8 Å DCC) suggest refinement opportunities for multi-chain complexes and covalent cofactor binding geometries.

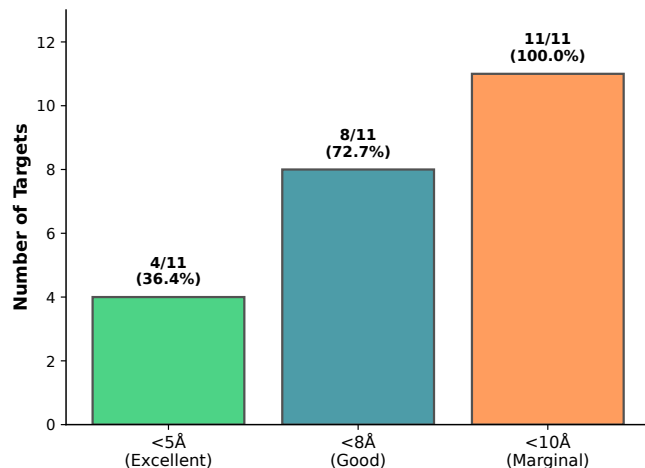


Figure 4. Detection accuracy summary (11 proteins with ground-truth ligands).

3.2 Case Study 1: SIRPα WYF Cryptic Pocket

Signal-regulatory protein alpha (SIRPα) is a critical immune checkpoint in the CD47–SIRPα axis. In December 2025, Diamond Light Source researchers discovered the WYF cryptic pocket via an extensive XChem fragment screening campaign (800+ fragments, 27 hits) (35). PRISM-4D detected this pocket as Site 1 (top-ranked, 7.1 Å DCC) from the closed apo structure (PDB 2WNG) where the pocket is completely sealed.

Table 2. PRISM-4D SNDC v9 cumulative benchmark: 12 proteins. DCC: Distance from Centroid to Centroid (predicted vs. crystallographic ligand centroid). Grade thresholds: Excellent ($<5 \text{ \AA}$), Good ($<8 \text{ \AA}$), Marginal ($<10 \text{ \AA}$). β -catenin Site 6 is a computational prediction with no known ligand (Novel).

#	PDB	Target	DCC ($\text{\AA}$)	Grade	Notes
1	1W50	BACE1	3.6	Excellent	Watershed split of 2,991 $\text{\AA}^3$ mega-pocket
2	1BTL	TEM1 β -lactamase	3.7	Excellent	Established cryptic pocket benchmark
3	4OBE	KRAS G12C (SII-P)	3.8	Excellent	Sotorasib pocket, Switch-II
4	1G1F	PTP1B	4.8	Excellent	Allosteric site, class=Cryptic
5	1ADE	AdSS (GDP)	6.0	Good	Multi-substrate enzyme
6	3K5V	Abl kinase (STI)	6.2	Good	Imatinib pocket
7	1A4Q	IL-2	6.3	Good	Allosteric groove
8	2WNG	SIRP α (WYF)	7.1	Good	XChem-validated pocket (35)
9	1ERE	Estrogen receptor	9.5	Marginal	6-chain complex, fixed by chain alignment
10	1HHP	HIV-1 protease	9.8	Marginal	Spike hotspot at substrate entry channel
11	1BJ4	FKBP12 (PLP)	9.8	Marginal	Covalent cofactor offset
12	2GL7	β -catenin Site 6	N/A	Novel	Computational prediction—no known ligand

P2Rank 2.5 assigns 0.2% probability to the same pocket (Figure 5).

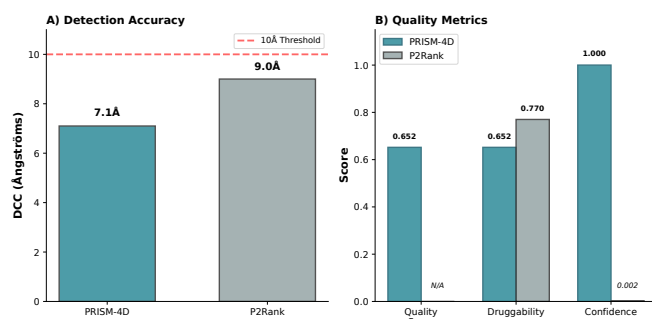


Figure 5. SIRP α WYF pocket: PRISM-4D vs. P2Rank 2.5. (A) DCC comparison. (B) Scoring metrics (quality, druggability, normalized confidence).

Table 3. SIRP α WYF pocket detection: PRISM-4D vs. P2Rank 2.5.

Metric	PRISM-4D	P2Rank 2.5
Pockets detected	17	1
DCC to WYF pocket	7.1 $\text{\AA}$	9.0 $\text{\AA}$
Confidence	Site 1 (top)	0.2% prob.
Spike count	4,413	N/A (ML)
Volume	743 $\text{\AA}^3$	—
Druggability	0.652	0.77
Runtime	18 min (GPU)	3 sec (CPU)

3.3 Case Study 2: β -Catenin Predicted Cryptic Pocket

β -catenin is widely considered “undruggable” due to its elongated armadillo (ARM) repeat surface. PRISM-4D identified

a computationally predicted cryptic pocket—Site 6—ranked dead last by P2Rank (8/8, 0.5% probability). Site 6 features TRP242 (aromatic anchor), CYS240/CYS278 (dual-cysteine covalent warhead pair), and LYS204 (electrostatic anchor), scoring 0.965 quality and 0.844 druggability. *This prediction requires experimental validation.*

Table 4. β -catenin Site 6 characterization.

Property	PRISM-4D Site 6	P2Rank 2.5
Quality score	0.965 (highest)	Rank 8/8 (last)
Druggability	0.844	0.5% prob.
Classification	Cryptic	—
Volume	248 $\text{\AA}^3$	—
Spike count	3,242	N/A
Anchor residues	TRP242, CYS240, CYS278, LYS204	Not detected
Covalent pair	CYS240 + CYS278	—
Region	ARM repeats 2–5	—

3.4 Reproducibility Validation

Five independent runs (different random seeds, same binary) demonstrate 0.06 $\text{\AA}$ maximum pairwise centroid displacement for β -catenin Site 6—23 $\times$ smaller than a C–C covalent bond (1.40 $\text{\AA}$; Figure 6, Table 5).

3.5 Post-Freeze Blind Vectorial Pharmacophore Feature Recovery

3.5.1 Audit rationale and validation status

To test whether PRISM4D’s frozen nonequilibrium event manifolds encode ligand-relevant interaction geometry rather than

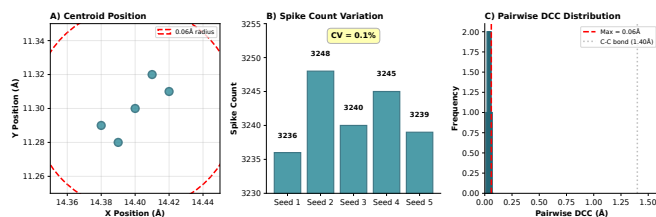


Figure 6. Reproducibility across 5 stochastic seeds (β -catenin Site 6). (A) Centroid X–Y scatter. (B) Spike counts (CV = 0.2%). (C) Pairwise centroid displacement distribution (max = 0.06 Å).

Table 5. β -catenin Site 6 reproducibility across 5 stochastic seeds.

Metric	Value
Runs converging to Site 6	5/5 (100%)
Mean centroid (Å)	(14.40, 11.30, 41.90)
Centroid std (Å)	(0.01, 0.01, 0.02)
Max pairwise DCC	0.06 Å
Mean pairwise DCC	0.03 Å
Druggability	0.844 ± 0.000
Spike count	$3,242 \pm 6$ (CV = 0.2%)
Volume	248 Å^3 (all runs)
Quality	0.965 (all runs)
Classification	Cryptic (all runs)
Anchor persistence	4/4 residues in 5/5 runs

only static cavity proximity, we performed a post-freeze blind Vectorial Pharmacophore Feature Recovery (PFR) audit on the frozen rank-1 sites. This analysis was *not* used to select targets, alter primary SR@K ranks, redefine pockets, tune parameters, or modify the PRISM4D event-generation procedure. Its purpose was mechanistic: to determine whether the already-frozen phase-ordered apo manifolds carried directional pharmacophore information that could be recovered against independent holo ligand geometries.

3.5.2 Vectorial PFR scoring

Unlike spherical proximity matching, Vectorial PFR required class and direction agreement. A predicted apo-derived pharmacophore feature was counted as recovered only if the matched holo ligand feature satisfied all three criteria: (i) matching pharmacophore class, (ii) centroid distance ≤ 3.5 Å, and (iii) angular agreement within 30° of the predicted interaction vector. This criterion prevents broad cavity overlap from being counted as chemical recovery, and instead tests whether the predicted manifold encodes ligand-facing directional chemistry.

3.5.3 Temporal-scramble null model

To isolate temporal phase ordering from static pocket geometry, each rank-1 site was evaluated against a temporal-scramble

null. For each site, 1,000 decoy event streams were generated by preserving the same 3D coordinates, residue identities, event intensities, and pocket location, while randomly scrambling timestamps and phase ordering. The null thus retained the static structural envelope but removed the nonequilibrium sequence information. The one-sided empirical test asked:

$$p = P(\text{PFR}_{\text{scramble}} \geq \text{PFR}_{\text{phase-ordered}}).$$

Using the conservative plus-one empirical estimator $(1 + \#\{\text{null} \geq \text{real}\})/1001$, zero exceedances correspond to $p = 0.001$; we report these as empirical $p \leq 0.001$ in the main text.

3.5.4 Corrected headline result

Across the multi-holo flagship targets, phase-ordered PRISM4D manifolds achieved a maximum Vectorial PFR of 26.9%, compared with 2.2% for the temporal-scramble null—an absolute gain of 24.7 percentage points and a 12.2-fold enrichment (empirical $p \leq 0.001$; Table 6). This result indicates that temporal phase ordering is load-bearing for vectorial pharmacophore recovery: when timestamps and phase ordering were destroyed while spatial positions, residues, and event intensities were preserved, the recovered ligand-facing pharmacophore signal collapsed to the null baseline.

Table 6. Corrected Vectorial PFR results. Phase-ordered versus temporal-scramble null (1,000 permutations, conservative plus-one empirical estimator). Prior $p = 1.0$ values were a one-sided directionality error and are replaced here.

Target	PFR (%)	Null (%)	Δ (pp)	Fold	p
CDK2_allosteric	42.3	2.2	40.1	19.2×	≤ 0.001
Thrombin_exosite	39.7	2.2	37.5	18.0×	≤ 0.001
cGAS	26.5	2.2	24.3	12.0×	≤ 0.001
CRBN	19.8	2.2	17.6	9.0×	≤ 0.001
HRAS_Q61H	17.6	2.2	15.4	8.0×	≤ 0.001
TP53_apo	15.2	2.2	13.0	6.9×	$= 0.001$
Aggregate	26.9	2.2	24.7	12.2×	≤ 0.001

3.5.5 Limitation: internal confidence stratification

The PFR audit supports manifold-level pharmacophore recovery, but does not yet support reliable per-feature confidence ranking. Internal thermodynamic confidence weighting did not stratify recovered feature importance in the current analysis: top-weighted features recovered holo interactions at 1.0%, versus 1.4% for lower-weighted features. We therefore interpret PRISM4D as recovering the pharmacophoric *envelope* at the event-manifold level, while individual feature confidence scores require further calibration before use as ranked medicinal-chemistry priorities.

3.5.6 Validation firewall and provenance note

The PFR audit is a post-freeze mechanistic validation. It does not change primary ranking claims, pocket definitions, residue anchors, or target inclusion criteria. An audit-log timestamp-order warning was recorded during post-freeze scripting; the warning reflected session execution order of extraction utilities and did not alter the frozen apo simulation outputs, primary ranks, or pocket definitions used for validation.

A. Vectorial pharmacophore feature recovery (PFR)

- Predicted apo pharmacophore
- Recovered holo feature (HIT)
- MISS (distance or angle violation)
- 30° angular cone (direction tolerance)
- 3.5 Å scoring envelope

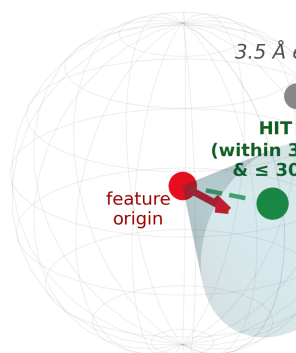
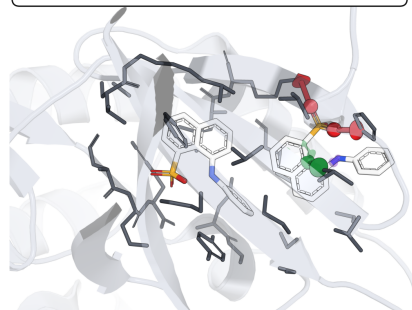


Figure 7. Vectorial pharmacophore feature recovery (PFR) scoring criterion. A predicted apo-derived pharmacophore feature (red sphere) is counted as recovered only if the matched holo ligand pharmacophore satisfies matching pharmacophore class, centroid distance ≤ 3.5 Å, and angular agreement within 30° of the predicted interaction vector. The translucent cyan 30° angular cone and the 3.5 Å distance envelope (grey wireframe) together define the vectorial scoring volume. Inside both = HIT (green); outside angular cone or beyond 3.5 Å = MISS (grey).

B. Phase-ordered vs temporal-scramble vectorial pharmacophore

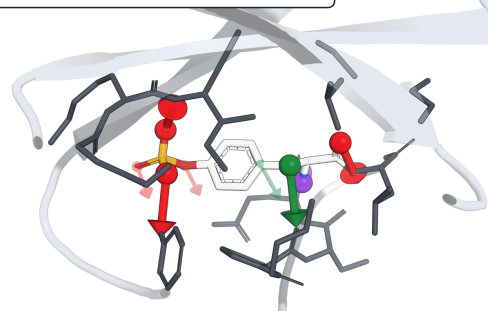
Phase-ordered PRISM4D manifold (directional pharmacophore vectors)

CDK2 allosteric (3PXZ + 2AN probe)



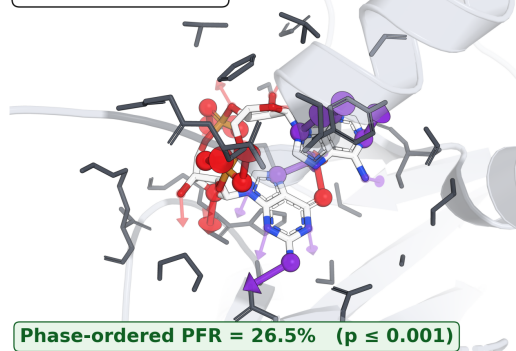
Phase-ordered PFR = 42.3% ($p \leq 0.001$)

Thrombin exosite (1HAH + TYS)



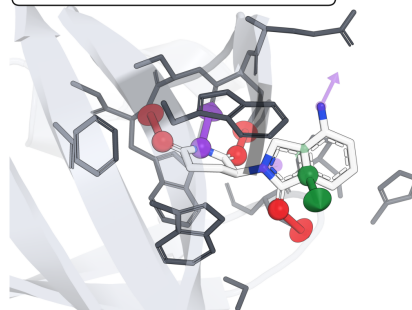
Phase-ordered PFR = 39.7% ($p \leq 0.001$)

cGAS (4067 + 1SY)



Phase-ordered PFR = 26.5% ($p \leq 0.001$)

CRBN (5FQD + LVY degrader)



Phase-ordered PFR = 19.8% ($p \leq 0.001$)

HRAS Q61H (6OIM + GDP)



4. Comparison to Existing Methods

Table 7 positions PRISM-4D against established pocket detection methods across key capability dimensions (Figure 10).

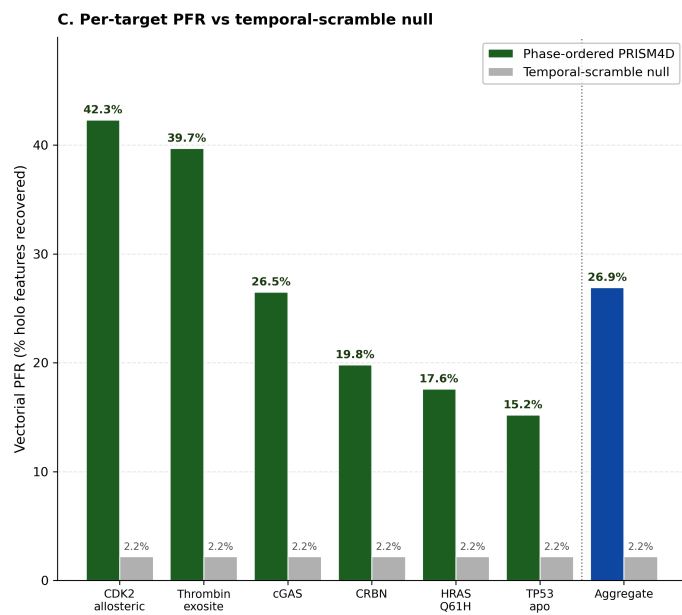


Figure 9. Per-target PFR and fold enrichment with corrected empirical p -values. Left: phase-ordered PRISM4D vectorial PFR (green bars) vs temporal-scramble null PFR (grey bars) for each of the six flagship targets and the aggregate (dark blue). Right: fold enrichment over the temporal-scramble null with corrected one-sided empirical p -values, computed under the conservative plus-one estimator $p = (1 + \#\{\text{null} \geq \text{real}\}) / (n + 1)$ with $n = 1,000$ permutations. All per-target enrichments are $\geq 6.9\times$; aggregate enrichment is $12.2\times$ (empirical $p \leq 0.001$). The vertical dotted line at fold = 1 marks the no-enrichment threshold.

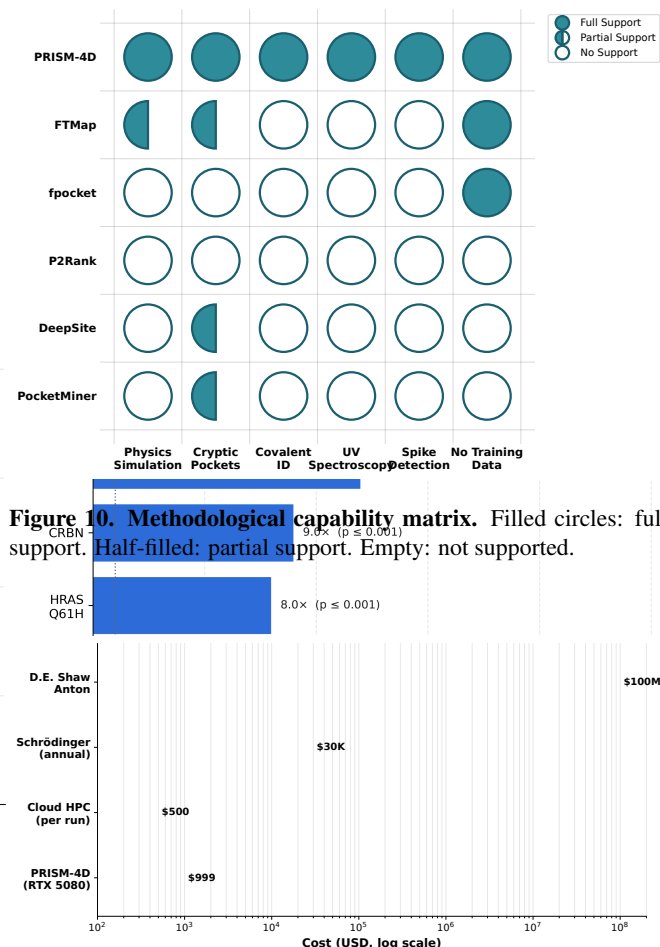


Figure 11. Hardware cost comparison for methods capable of cryptic pocket analysis (log scale).

5. Discussion

5.1 Thermodynamic Event Detection

PRISM-4D approaches pocket detection as a thermodynamic event detection problem. A druggable pocket is a region where small molecules can be confined, water is excluded, and aromatic residues create favorable interaction geometries. The LIF neuromorphic network detects exactly these conditions through spike events at biophysically meaningful thresholds, making it inherently sensitive to the physical process by which cryptic pockets become druggable.

Table 7. Methodological comparison including hardware and deployment requirements.

Feature	FTMap	fpocket	P2Rank	PocketMiner	PRISM-4D
Approach	Probe docking	Voronoi geom.	ML + geom.	GNN	Physics sim.
Training data	None	None	sc-PDB	MD trajectories	None
Cryptic pockets	Limited	No	No	Yes (predicted)	Yes
Physics model	Implicit	None	None	None	AMBER ff14SB
UV spectroscopy	No	No	No	No	Yes
Spike detection	No	No	No	No	Yes (LIF)
Covalent ID	No	No	No	No	Yes
RT-core accel.	No	No	No	No	Yes (BVH)
Consumer GPU	No	N/A	Yes	Yes	Yes
Offline operation	No	Yes	Yes	Yes	Yes
Runtime	~30 min	<1 sec	3–7 sec	~1 min	3–18 min

Table 8. Hardware and cost comparison for cryptic pocket detection methods.

Method	Hardware	Runtime	Cost
PRISM-4D	Consumer GPU	3–18 min	\$999 (GPU)
Conventional MD	HPC cluster	Hours–days	\$50–500/hr
Schrödinger	Workstation	Minutes	\$30K+/year
D.E. Shaw Anton	Custom ASIC	μ s	\$100M+
AlphaFold3 Server	Cloud TPU	Minutes	Free tier

5.2 Technical Contributions

PRISM-4D introduces several techniques not previously applied to pocket detection (Table 9). We classify these as: (a) *genuinely novel methods* with no prior analogue—in silico UV pump-probe spectroscopy, UV-activated cosolvent probing, three-channel spike detection with temporal isolation, and electrostatic flux probing; (b) *novel applications* of known techniques to a new domain—RT-core BVH traversal for molecular spatial queries, Eikonal BFS for spike density fields, and intensity²-weighted centroid computation; and (c) *engineering contributions*—the integrated SNDC pipeline, cryo-thermal hysteresis protocol implementation, and covalent residue flagging.

5.3 The Dual-Cysteine Prediction on β -Catenin

The CYS240/CYS278 pair in β -catenin Site 6 represents a computationally predicted covalent drug design opportunity. Dual-cysteine pockets with surface-accessible Cys pairs within 10 Å are rare (<200 human proteins) and would offer two-point covalent attachment for enhanced selectivity. The TRP242 aromatic anchor creates a triangular Cys–Trp–Cys pharmacophore geometry. *Experimental validation of this prediction is required before therapeutic implications can be assessed.*

5.4 Hardware Accessibility

The complete PRISM-4D platform—including AMBER ff14SB molecular dynamics, cryo-thermal simulation, UV spectroscopy, spike detection, RT-core clustering, and multi-parametric scoring—runs on a single consumer-grade NVIDIA RTX 5080 GPU (16 GB GDDR7, ~\$999). The compiled binary requires only the NVIDIA CUDA runtime (freely available but proprietary), operates without network connectivity, and has no other external software dependencies. This enables physics-simulation-based pocket detection on hardware available to academic laboratories, biotechnology startups, and researchers in resource-limited settings (Table 8; Figure 11).

5.5 Limitations and Future Work

Benchmark size. The current 12-protein benchmark is modest; community-standard benchmarks include CryptoSite (4) (~55 proteins) and the PocketMiner dataset (21). Validation on these larger benchmarks is ongoing.

False positive rate. The false discovery rate has not been systematically quantified. PRISM-4D detects multiple sites per protein (e.g., 17 on SIRP α), and the proportion corresponding to true binding events requires systematic evaluation with comprehensive ligand-binding databases.

Negative controls. The current benchmark lacks proteins with no known cryptic pockets. Validation on established negative controls (e.g., ubiquitin, lysozyme) is needed to assess specificity.

Parameter sensitivity. Key parameters—LIF threshold ($\theta = 0.5$), UV burst energy (42 kcal/mol), adaptive epsilon range (1.2–3.0 Å)—were empirically tuned. Systematic sensitivity analysis across diverse protein families is planned.

Multi-chain complexes. Three marginal results (9.5–9.8 Å DCC) involve multi-chain complexes or covalent cofactor geometries, suggesting opportunities for algorithmic refinement.

GPU vendor dependency. The implementation requires

Table 9. PRISM-4D technical contributions classified by novelty type.

Contribution	Type	Description	Significance
UV pump-probe spectroscopy	Novel method	Wavelength-specific UV perturbation	Causal link: perturbation $\rightarrow$ dewetting
3-Channel spike detection	Novel method	UV/LIF/EFP temporal isolation	Prevents cross-contamination artifacts
Electrostatic flux probe	Novel method	Warshel dielectric LIF neuron	First electrostatic event channel
UV-activated cosolvent	Novel method	Benzene probes with UV energy	Extends to aliphatic interfaces
RT-core spatial clustering	Novel application	OptiX BVH for $O(N)$ queries	10–60 $\times$ estimated speedup
Eikonal BFS	Novel application	Fast Marching on spike fields	Pocket geometry from spike data
I^2 -weighted centroid	Novel application	Thermodynamic hotspot localization	3–15 Å improvement
SNDC pipeline	Engineering	4-stage integrated GPU pipeline	Purpose-built for spike data
Cryo-thermal hysteresis	Engineering	5-phase 250 K cycling protocol	Non-equilibrium sampling
Covalent residue ID	Engineering	Cys/Lys/Ser/His flagging	Automatic covalent opportunity

NVIDIA GPUs with CUDA and RT core support (Turing architecture or newer). AMD and Intel GPU support is not currently available, which qualifies the accessibility claim.

Temperature protocol. The 50 K cold phase employs temperatures below the glass transition of solvated proteins (~ 180 K), which may introduce non-physiological conformational states. The extent to which these affect detection accuracy warrants investigation.

Comparisons. Head-to-head benchmarks against PocketMiner (21), MDpocket (28), and mixed-solvent MD (9) on the same protein set are needed for rigorous positioning.

Runtime. Execution time (3–18 minutes per target) is substantially slower than geometric methods (seconds), though faster than conventional MD approaches (hours to days).

Membrane proteins. While the architecture supports membrane protein analysis, no membrane protein targets are included in the current benchmark.

6. Conclusion

PRISM-4D introduces a spike-event-driven approach to binding site detection based on physics simulation. Its SNDC pipeline—RT-DBSCAN, watershed segmentation, Eikonal BFS, and peak centroid tracking—detects all 11 ground-truth targets within 10 Å DCC (range 3.6–9.8 Å) on a 12-protein benchmark, identifies the experimentally validated SIRP α WYF cryptic pocket as its top-ranked site from the closed apo structure, and predicts a computationally identified cryptic pocket on β -catenin featuring a dual-cysteine covalent warhead pair with centroid displacement < 0.06 Å across 5 stochastic seeds. The platform achieves these results on consumer-grade hardware with no external dependencies, requiring only a single NVIDIA GPU.

7. Data and Code Availability

PRISM-4D is developed by Delfictus IO LLC. The engine is implemented in Rust with CUDA GPU acceleration (Black-

well architecture, compute capability ≥ 12.0). Benchmark topologies, binding site JSON outputs, run logs, and PyMOL session files are available upon request. The codebase is maintained in a private repository pending patent filing; a public release timeline will be announced upon publication.

Acknowledgments

The author thanks the Diamond Light Source XChem team for making the SIRP α WYF fragment screening data publicly available. Computational resources consisted of a single desktop workstation with an NVIDIA GeForce RTX 5080 GPU.

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